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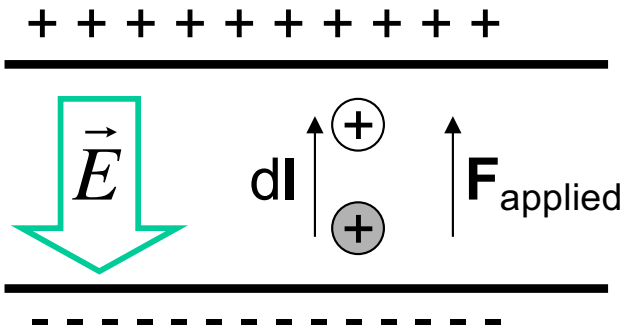
Lectures 5

Potential Functions for Static Fields

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Electric Potential



How much potential energy does a charge have in an electric field?

Answer: It depends on the work that you had to do to get it to that spot. You have to work against the E-field.

$$\Delta PE = \vec{F}_{applied} \bullet d\vec{l} = -\vec{F}_e \bullet d\vec{l}$$

$$\vec{F}_e = q\vec{E}$$

$$\Delta PE = -q\vec{E} \bullet d\vec{l}$$

Answer depends on the amount of charge q
Can we define something determined by the field only

Analogy to gravity: $\vec{F}_g = m\vec{g},\ PE = mgh,\ \dfrac{PE}{m} = gh$

Electric Potential - Definition

Work you do per unit charge is the ELECTRIC POTENTIAL DIFFERENCE between the two points

$$\Delta V = \frac{\Delta PE}{q} = -\vec{E} \bullet d\vec{l} = -E_x \Delta x \ \ if \ d\vec{l} = \Delta x \hat{a}_x$$

Units: $\frac{Nm}{C} = \frac{J}{C} = VOLTS$

$$E_x = -\frac{\Delta V}{\Delta x}, E_y = -\frac{\Delta V}{\Delta y}, E_z = -\frac{\Delta V}{\Delta z}$$ Units: $\frac{V}{m}$

$$\vec{E} = -\nabla V$$

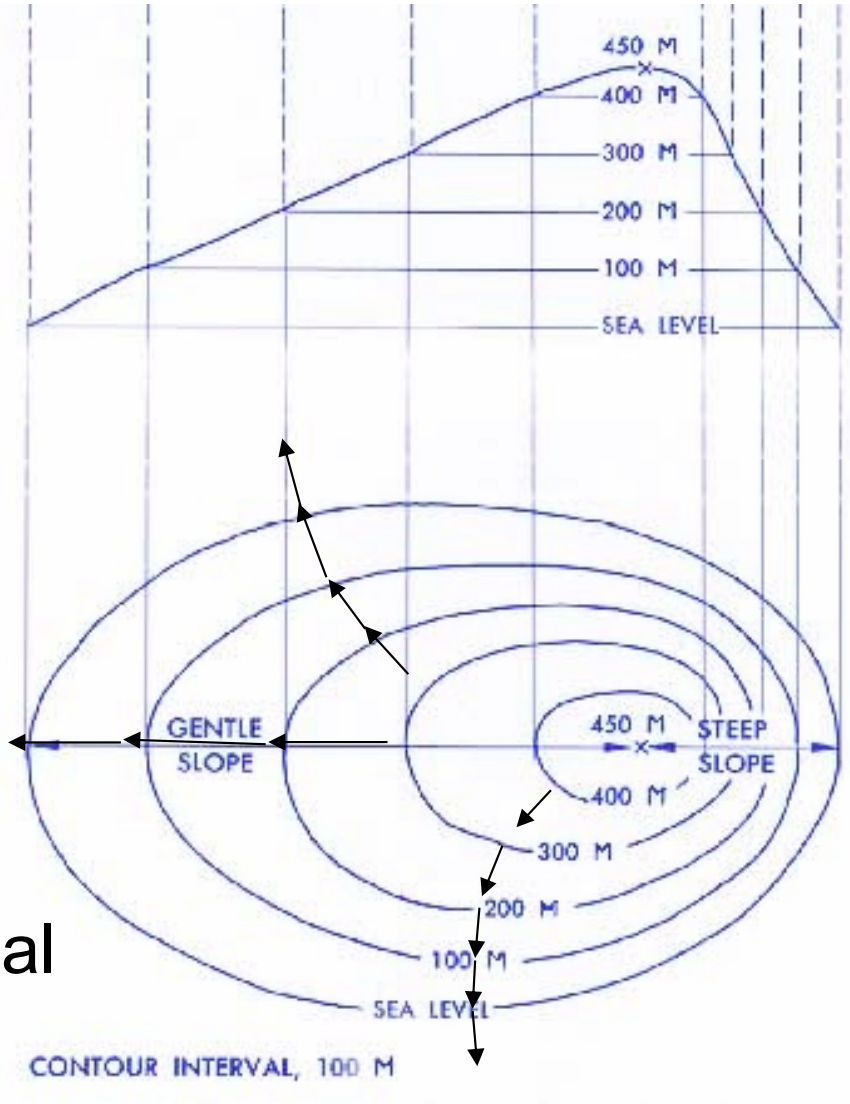
Gradient of E $\nabla V \equiv \left(\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z}\right)$

Potential: Mount Electron



What is the steepest slope on the mountain?

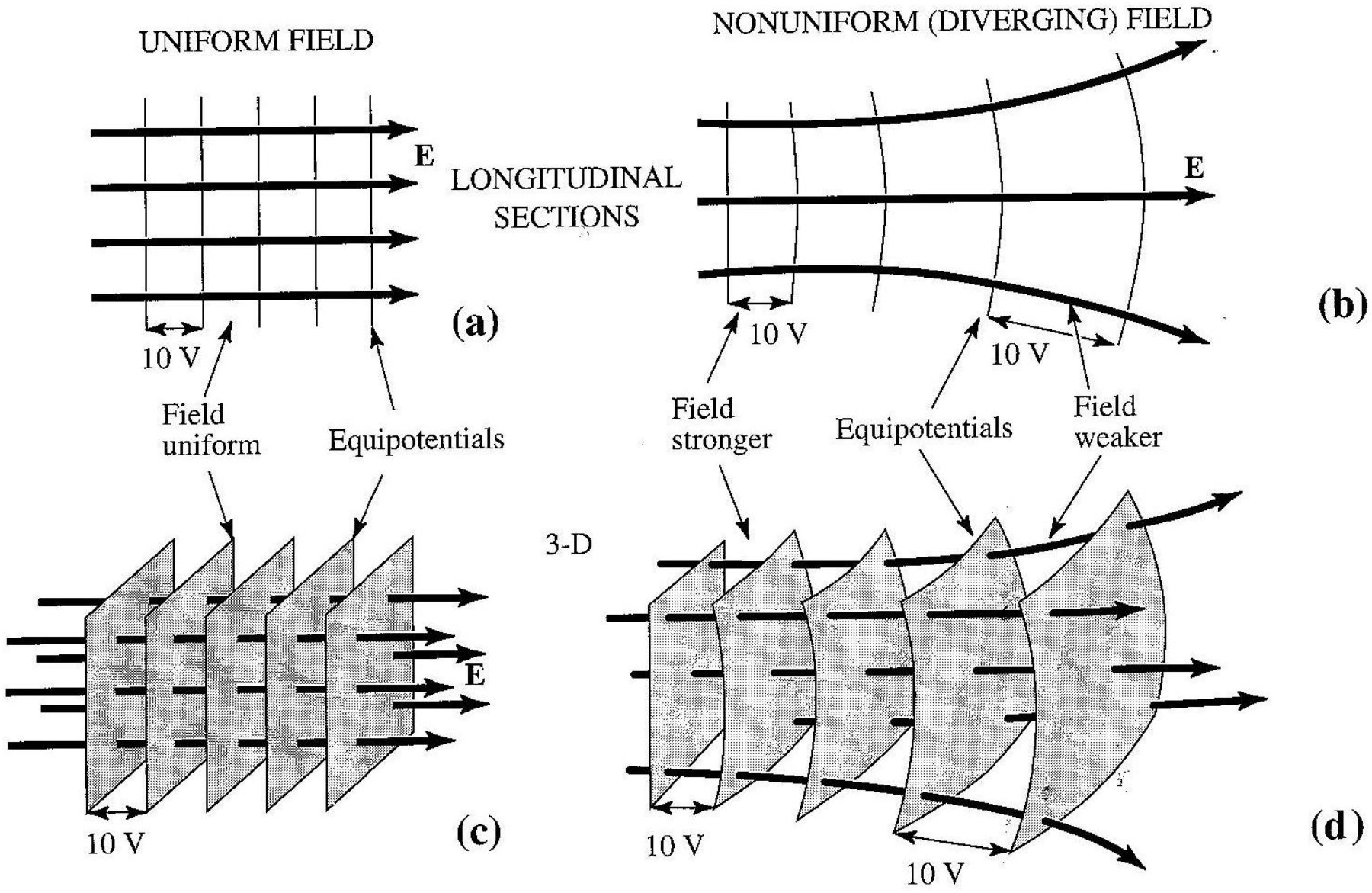
Potential



Surfaces of equal potential

What direction is the steepest slope at a given point?

Ans: The steepest downslope is ***opposite*** *the gradient! This is always perpendicular to surfaces of constant potential*


UNIFORM FIELD
NONUNIFORM (DIVERGING) FIELD
E
LONGITUDINAL SECTIONS
E
(a)
(b)
10 V
10 V
10 V
Field uniform
Equipotentials
Field stronger
Equipotentials
Field weaker
3-D
E
(c)
(d)
10 V
10 V
10 V

Gradient Definition

$$\vec{E} = -\left(\frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z\right)$$

Most important definition of today:

$$\vec{E} = -\nabla V$$ V: electrostatic potential

$$\nabla = \hat{a}_x \frac{\partial}{\partial x} + \hat{a}_y \frac{\partial}{\partial y} + \hat{a}_z \frac{\partial}{\partial z}$$

Or $$\nabla = \left(\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z}\right)$$

Del **Operator** (Cartesian Coordinates)
used to do gradient, divergence, and curl

Scalar Potentials

- For: $\Phi_1(x,y,z) = x^2 + y^2 + z^2$

  $\Phi_2(x,y,z) = x + 2y + 2z$

Find the following quantities at (3, 4, 12):

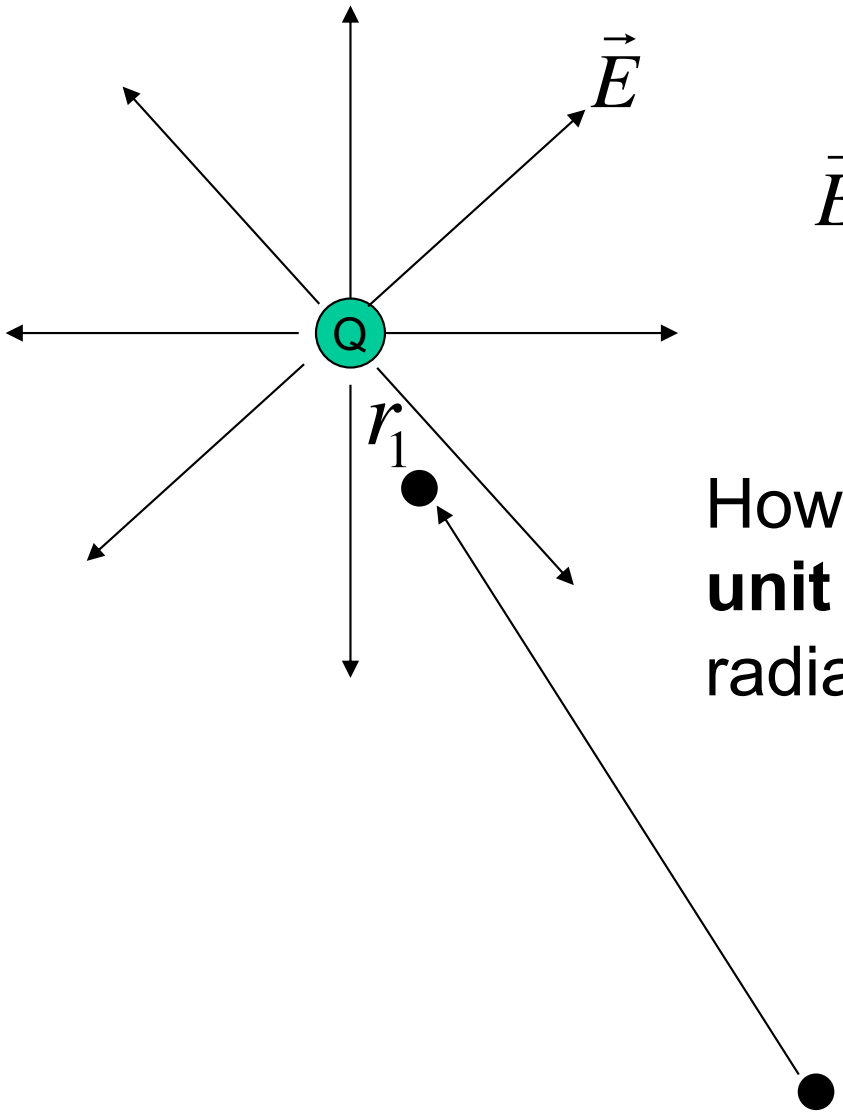
(a) the maximum rate of increase of Φ_1

(b) the maximum rate of increase of Φ_2

(c) the rate of increase of Φ_1 along the direction of the maximum rate of increase of Φ_2

From D5.2 (p 290) of old book

Example: Potentials for a Point Charge

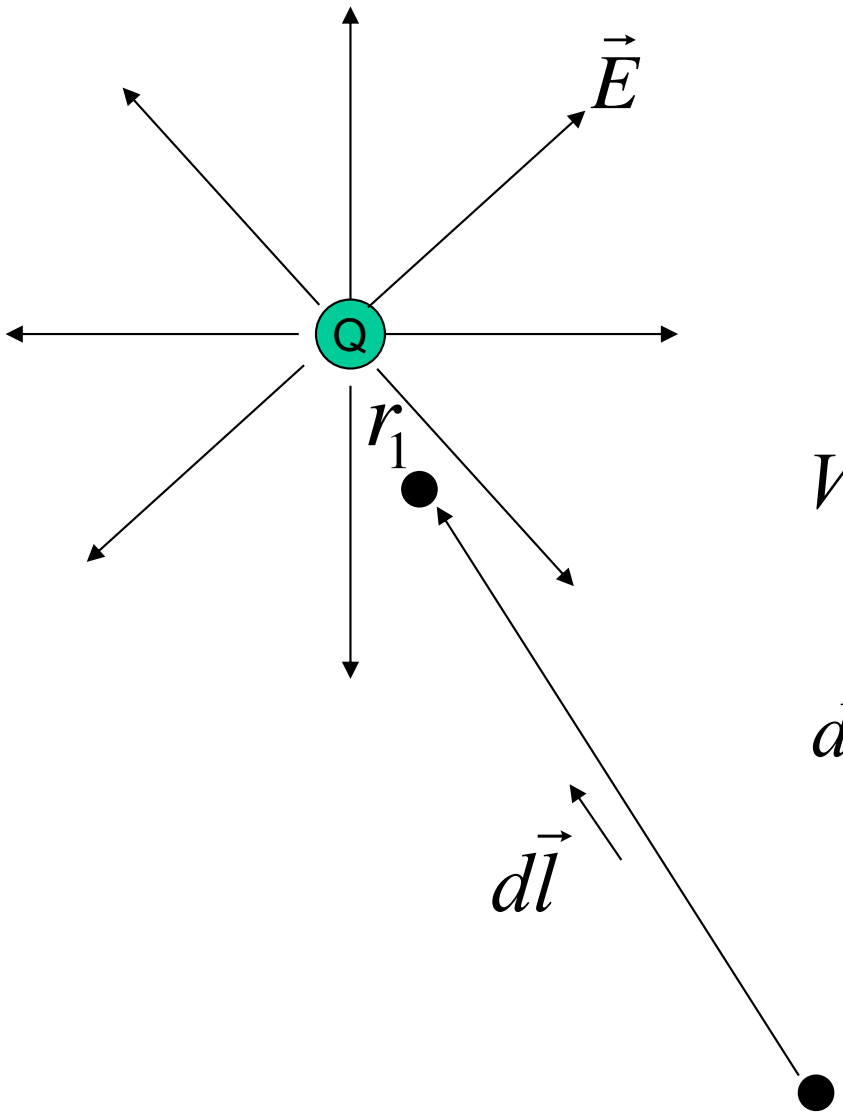


$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{a}_r$$

In spherical coords

How much work is required to move a **unit** charge from infinity to a radial distance = r_1

Example: Potentials for a Point Charge



$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{a}_r$$ In spherical coords

$$\vec{E} = -\nabla V$$

$$V(r) = -\int_{\infty}^{r_1} \vec{E} \bullet d\vec{l}$$

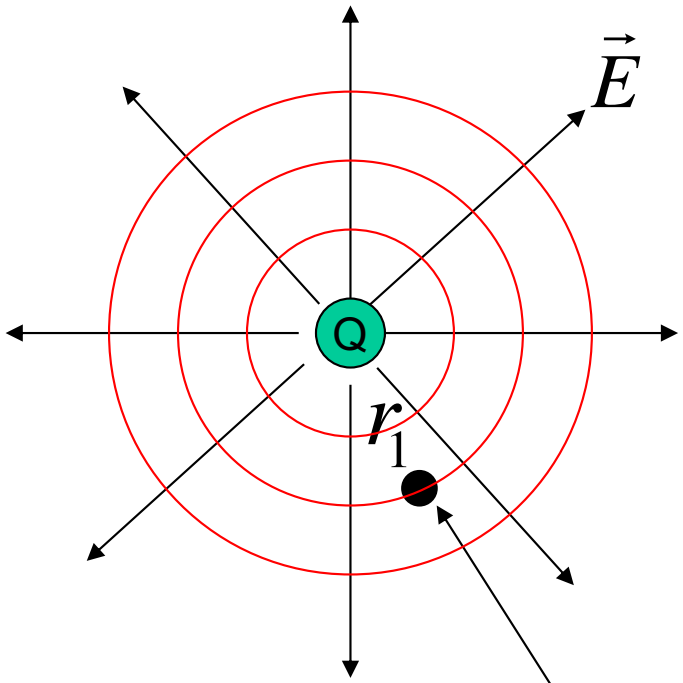
"Absolute" potential at r_1 using zero potential at r = ∞

$$d\vec{l} = -|dr|\hat{a}_r = dr\hat{a}_r$$

Since dr<0 going from r=∞

$$V(r) = -\int_{\infty}^{r_1} \frac{Q}{4\pi\varepsilon_0 r^2} dr$$

Example: Potentials for a Point Charge



Surfaces of constant potential are spheres in 3D
-same amount of work

$$V(r) = -\int_{\infty}^{r_1} \frac{Q}{4\pi\varepsilon_0 r^2}\, dr$$

$$= \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_1} - \frac{1}{\infty}\right)$$

$$= \frac{Q}{4\pi\varepsilon_0 r_1}$$

Challenge Question: Scalar Potentials

- If the scalar potential V can be defined, which of the following is true?

(a) The potential difference between two points is identical, independent of path

(b) The electric field is conservative

(c) The electric field lines will always be perpendicular to equipotential surfaces

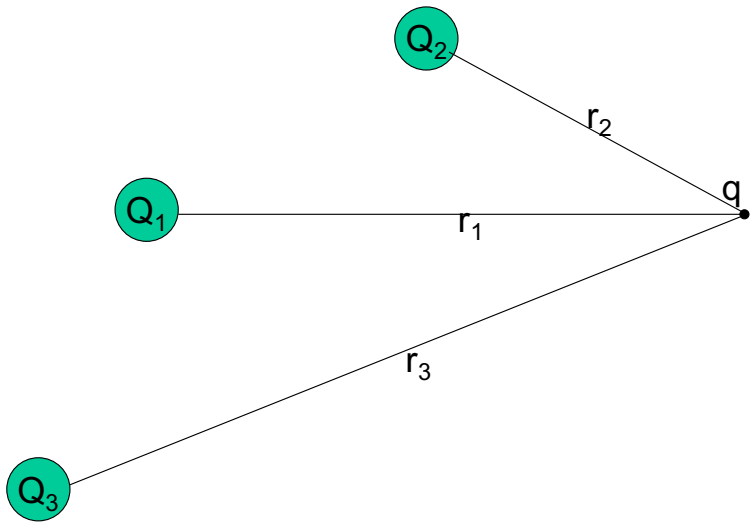
(d) The greatest decrease in potential per unit length for a positive charge is along the electric field direction

(e) All of the above

Potential Superposition

Potentials of more than one point charge are superimposed by addition

$$V_{\text{point}} = \frac{1}{4\pi\varepsilon_0}\left(\frac{Q_1}{r_1} + \frac{Q_2}{r_2} + \frac{Q_3}{r_3} + \bullet\bullet\bullet\right)$$



Scalar Potentials

- For: $\vec{E} = yz\hat{a}_x + (y + zx)\hat{a}_y + xy\hat{a}_z$

(a) Find the scalar potential if V(0,0,0)=0. (Hint: Use a direct line path. At home, try using 3 separate segments along x, y, and z directions.)
(b) Evaluate the potential difference V_A-V_B for:

1. A=(2,1,1) and B=(1,4,0.5)
2. A=(2,2,2) and B=(1,1,1)
3. A=(5,1,0.2) and B=(1,2,3)

Adapted from D 5.4 (p299) of old book

Useful facts about Potential

- Potential is always the DIFFERENCE between two points
- If one point is at infinity, then the potential is an absolute potential
- The gradient of V gives the vector -**E** at a particular point
- Potential between two points is identical, regardless of whether a straight or curved path is taken
  - **E** is a conservative field
- Electric field lines are always perpendicular to equipotential surfaces (constant voltage)
  - E-field lines are along the direction of the greatest decrease in potential

Gradient Function in Cylindrical and Spherical Coordinates

Φ = scalar field (like voltage)

$$\nabla\Phi = \begin{cases} \left(\frac{\partial\Phi}{\partial x}\hat{a}_x + \frac{\partial\Phi}{\partial y}\hat{a}_y + \frac{\partial\Phi}{\partial z}\hat{a}_z\right) & \text{Cartesian} \\ \left(\frac{\partial\Phi}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial\Phi}{\partial\phi}\hat{a}_\phi + \frac{\partial\Phi}{\partial z}\hat{a}_z\right) & \text{Cylindrical} \\ \left(\frac{\partial\Phi}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial\Phi}{\partial\theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial\Phi}{\partial\phi}\hat{a}_\phi\right) & \text{Spherical} \end{cases}$$

Lecture 5 Summary

- Since $\nabla \cdot \vec{\boldsymbol{D}} = \rho$ and $\vec{\boldsymbol{D}} = \epsilon_o \vec{\boldsymbol{E}}$, use vector identity $\nabla \cdot \nabla \Phi = 0$

$$\vec{E} = -\nabla \Phi$$

satisfy Gauss' Laws and is valid for static fields